

Note 12: Discrete error theorem, fault torrent computation

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Date: 2026.7.10

12.1 Summary of Progress

1. Establish complete QECC correction conditions and error discretization theorem with rigorous proofs.
2. Systematically derive diamond norm definitions, core properties and composite noise bound for channel error quantification.
3. State and fully prove the concatenated quantum fault-tolerance threshold theorem based on quadratic error recursion.

12.2 The proof for discrete error theorem

Notation: $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$.

Theorem 1 (Quantum Error Correction Condition) Given Kraus operators $\mathcal{E} = \{E_k\}_{k=1}^m$ and code subspace $\mathcal{C}$. Assume

- Trace preserving: $\sum_k E_k^\dagger E_k = I$;
- The orthogonal projector $P : \mathcal{H} \rightarrow \mathcal{C}$ satisfies $PE_i^\dagger E_j P = \alpha_{ij} P$ for all i, j ;
- The coefficient matrix $\alpha = (\alpha_{ij})$ is Hermitian, i.e. $\alpha_{ij} = \overline{\alpha_{ji}}$.

Then for the noise channel

$$\mathcal{E}(\rho) := \sum_{k=1}^m E_k \rho E_k^\dagger, \quad \forall \rho \in \mathcal{C},$$

there exists a trace-preserving quantum recovery operation $\mathcal{R}$ such that $\mathcal{R}(\mathcal{E}(\rho)) \propto \rho$ for all $\rho = P\rho P$.

Proof Since α is Hermitian matrix, we perform unitary diagonalization: there exists unitary matrix u and real diagonal matrix $d = \text{diag}(d_{11}, \dots, d_{mm})$, such that

$$\alpha_{ij} = \sum_t u_{it} d_{tt} \overline{u_{jt}}.$$

Define rotated Kraus operators $F_l = \sum_i u_{il} E_i$. Then

$$\begin{aligned} \sum_l F_l \rho F_l^\dagger &= \sum_{l,k,j} (u_{lj} E_j) \rho (u_{lk} E_k)^\dagger \\ &= \sum_{l,k,j} u_{lj} \overline{u_{kl}} E_j \rho E_k^\dagger = \sum_{k,j} \delta_{jk} E_j \rho E_k^\dagger \\ &= \sum_k E_k \rho E_k^\dagger = \mathcal{E}(\rho). \end{aligned}$$

Substitute into the error correction condition:

$$PF_k^\dagger F_l P = \sum_{i,j} \overline{u_{ki}} u_{jl} PE_i^\dagger E_j P = \sum_{i,j} \overline{u_{ki}} u_{jl} \alpha_{ij} P = d_{kl} P.$$

By polar decomposition,

$$F_k P = U_k \sqrt{(F_k P)^\dagger F_k P} = U_k \sqrt{P F_k^\dagger F_k P} = \sqrt{d_{kk}} U_k P$$

for some unitary U_k .

Construct recovery operation

$$\mathcal{R}(\sigma) = \sum_k P U_k^\dagger \sigma U_k P.$$

For arbitrary code state $\rho = P \rho P$, direct calculation

$$\begin{aligned} P U_k^\dagger F_l \rho F_l^\dagger U_k P &= \frac{(F_k P)^\dagger}{\sqrt{d_{kk}}} F_l \rho F_l^\dagger \frac{F_k P}{\sqrt{d_{kk}}} \\ &= \frac{1}{d_{kk}} P F_k^\dagger F_l \rho F_l^\dagger F_k P = \frac{1}{d_{kk}} P F_k^\dagger F_l P \rho P F_l^\dagger F_k P \\ &= \frac{1}{d_{kl}} d_{kl} d_{kk} P \rho P = \delta_{kl} d_{kk} \rho. \end{aligned}$$

Therefore

$$\begin{aligned} \mathcal{R}(\mathcal{E}(\rho)) &= \sum_{k,l} P U_k^\dagger F_l \rho F_l^\dagger U_k P \\ &= \sum_{k,l} \delta_{kl} d_{kk} \rho = \left(\sum_k d_{kk} \right) \rho \propto \rho. \end{aligned}$$

The existence of recovery operation $\mathcal{R}$ is verified. ■

Recover operation:

$$\mathcal{R}(\sigma) = \sum_k P U_k^\dagger \sigma U_k P.$$

Theorem 2 (Discretization of the errors) Let $\mathcal{E} = \{E_k\}_{k=1}^m$ satisfy the condition in Theorem 1.

Suppose another quantum operation $\tilde{\mathcal{E}}$ has Kraus operators $\{\tilde{E}_j\}$ which are linear combinations of $\{E_i\}$, i.e.

$$\tilde{E}_j = \sum_i m_{ji} E_i$$

for some $m_{ij} \in \mathbb{C}$. Then the same recovery operation $\mathcal{R}$ can also correct the noise process $\tilde{\mathcal{E}}$ on code C .

Proof Since $F_l = \sum_i u_{il} E_i$ and $u = (u_{il})$ is unitary, we have $u^{-1} = (u^{il})$. Then

$$(E_i)_{1 \leq i \leq m} = u^{-1} (F_l)_{1 \leq l \leq m} = (u^{il})_{1 \leq i, l \leq m} (F_l)_{1 \leq l \leq m}.$$

Substitute the linear relation $\tilde{E}_j = \sum_i m_{ji} E_i$, then

$$\tilde{E}_j = \sum_i m_{ji} E_i = \sum_{i,l} m_{ji} u^{il} F_l = \sum_l c_{jl} F_l.$$

Use the conclusion

$$P U_k^\dagger F_l \rho F_l^\dagger U_k P = \delta_{kl} d_{kk} \rho$$

to compute the recovery channel acting on the noisy state $\tilde{\mathcal{E}}(\rho) = \sum_j \tilde{E}_j \rho \tilde{E}_j^\dagger$:

$$\begin{aligned}
\mathcal{R}(\tilde{\mathcal{E}}(\rho)) &= \mathcal{R}\left(\sum_i \tilde{E}_i \rho \tilde{E}_i^\dagger\right) \\
&= \sum_{k,i} P U_k^\dagger \left(\tilde{E}_i \rho \tilde{E}_i^\dagger \right) U_k P \\
&= \sum_{k,i,j} P U_k^\dagger \left(c_{ij} F_j \rho \overline{c_{ij}} F_j^\dagger \right) U_k P \\
&= \sum_{k,i,j} |c_{ij}|^2 P U_k^\dagger \left(F_j \rho F_j^\dagger \right) U_k P \\
&= \sum_{k,i,j} |c_{ij}|^2 \delta_{kj} d_{kk} \rho \propto \rho.
\end{aligned}$$

This completes the proof that $\mathcal{R}$ also corrects the noise process $\mathcal{F}$. ■

12.3 Introduction to Fault torance computation

The norm of channels

Definition 3 (Diamond Norm) Let $\mathcal{M}(\mathbb{C}^d)$ be the d -dimensional matrix algebra, and let $\mathcal{L} : \mathcal{M}(\mathbb{C}^a) \rightarrow \mathcal{M}(\mathbb{C}^b)$ be a linear superoperator. The diamond norm of $\mathcal{L}$ is defined as

$$\|\mathcal{L}\|_\diamond = \sup_{n \in \mathbb{N}} \|\mathcal{I}_n \otimes \mathcal{L}\|_{1 \rightarrow 1},$$

where the trace operator norm is

$$\|\mathcal{T}\|_{1 \rightarrow 1} = \sup_{\rho \geq 0, \text{tr } \rho \leq 1} \|\mathcal{T}(\rho)\|_1, \quad \|X\|_1 = \text{tr} \sqrt{X^\dagger X}.$$

For quantum channels $\mathcal{N}_1, \mathcal{N}_2$, the diamond distance $\|\mathcal{N}_1 - \mathcal{N}_2\|_\diamond$ quantifies the worst-case distinguishability over entangled joint systems.

Proof that $\|\cdot\|_\diamond$ satisfies triangle inequality (norm axiom): Take two linear superoperators $\mathcal{L}_1, \mathcal{L}_2$. For any fixed $n \in \mathbb{N}$,

$$\|\mathcal{I}_n \otimes (\mathcal{L}_1 + \mathcal{L}_2)\|_{1 \rightarrow 1} = \|\mathcal{I}_n \mathcal{L}_1 + \mathcal{I}_n \mathcal{L}_2\|_{1 \rightarrow 1} \leq \|\mathcal{I}_n \otimes \mathcal{L}_1\|_{1 \rightarrow 1} + \|\mathcal{I}_n \otimes \mathcal{L}_2\|_{1 \rightarrow 1},$$

where we used the triangle inequality of trace operator norm $\|\cdot\|_{1 \rightarrow 1}$. Taking supremum over all n on both sides:

$$\|\mathcal{L}_1 + \mathcal{L}_2\|_\diamond \leq \sup_n \|\mathcal{I}_n \otimes \mathcal{L}_1\|_{1 \rightarrow 1} + \sup_n \|\mathcal{I}_n \otimes \mathcal{L}_2\|_{1 \rightarrow 1} = \|\mathcal{L}_1\|_\diamond + \|\mathcal{L}_2\|_\diamond.$$

The other norm axioms (non-negativity, homogeneity) follow trivially from the trace norms properties.

Lemma 4 (Fundamental Estimation Properties of Diamond Norm) Let $\mathcal{L}, \mathcal{L}_1, \mathcal{L}_2$ be linear superoperators, $\mathcal{I}$ identity single-qubit channel, $\mathcal{I}_n$ identity map on n -qubit space.

1. **Tensor invariance:** $\|\mathcal{I}_n \otimes \mathcal{L}\|_\diamond = \|\mathcal{L}\|_\diamond$;
2. **Multiplicativity under composition:** $\|\mathcal{L}_1 \circ \mathcal{L}_2\|_\diamond \leq \|\mathcal{L}_1\|_\diamond \cdot \|\mathcal{L}_2\|_\diamond$;
3. **CPTP channel bound:** If $\mathcal{N}$ is CPTP quantum channel, then $\|\mathcal{N}\|_\diamond = 1$.

Proof of each property:

1. Tensor invariance: By definition $\|\mathcal{I}_n \otimes \mathcal{L}\|_\diamond = \sup_m \|\mathcal{I}_m \otimes (\mathcal{I}_n \otimes \mathcal{L})\|_{1 \rightarrow 1} = \sup_{mn} \|\mathcal{I}_{mn} \otimes \mathcal{L}\|_{1 \rightarrow 1} = \sup_k \|\mathcal{I}_k \otimes \mathcal{L}\|_{1 \rightarrow 1} = \|\mathcal{L}\|_\diamond$.

2. Composition multiplicativity:

$$\|\mathcal{I}_k \otimes (\mathcal{L}_1 \circ \mathcal{L}_2)\|_{1 \rightarrow 1} = \|(\mathcal{I}_k \otimes \mathcal{L}_1) \circ (\mathcal{I}_k \otimes \mathcal{L}_2)\|_{1 \rightarrow 1} \leq \|\mathcal{I}_k \otimes \mathcal{L}_1\|_{1 \rightarrow 1} \cdot \|\mathcal{I}_k \otimes \mathcal{L}_2\|_{1 \rightarrow 1}.$$

Take supremum over k , the inequality holds for diamond norm.

3. CPTP norm equal to 1: For any CPTP $\mathcal{N}$, $\mathcal{I}_k \otimes \mathcal{N}$ is also CPTP, and trace norm of CPTP map is bounded by 1. There exists maximally entangled state achieving equality, so $\|\mathcal{N}\|_\diamond = 1$.

Theorem 5 (Composite Diamond Norm Error Bound) Let $\mathcal{N}_1, \mathcal{N}_2$ be CPTP maps on $\mathcal{M}(\mathcal{H})$, $\mathcal{I}$ the identity map. Denote $\varepsilon_1 = \|\mathcal{N}_1 - \mathcal{I}\|_\diamond$, $\varepsilon_2 = \|\mathcal{N}_2 - \mathcal{I}\|_\diamond$. Then

$$\|\mathcal{N}_1 \circ \mathcal{N}_2 - \mathcal{I}\|_\diamond \leq \varepsilon_1 + \varepsilon_2 + \varepsilon_1 \varepsilon_2.$$

Proof . Algebraic decomposition:

$$\mathcal{N}_1 \circ \mathcal{N}_2 - \mathcal{I} = (\mathcal{N}_1 - \mathcal{I}) \circ (\mathcal{N}_2 - \mathcal{I}) + (\mathcal{N}_1 - \mathcal{I}) + (\mathcal{N}_2 - \mathcal{I}).$$

By triangle inequality of diamond norm:

$$\|\mathcal{N}_1 \circ \mathcal{N}_2 - \mathcal{I}\|_\diamond \leq \|(\mathcal{N}_1 - \mathcal{I}) \circ (\mathcal{N}_2 - \mathcal{I})\|_\diamond + \|\mathcal{N}_1 - \mathcal{I}\|_\diamond + \|\mathcal{N}_2 - \mathcal{I}\|_\diamond.$$

Use multiplicative property $\|\mathcal{A} \circ \mathcal{B}\|_\diamond \leq \|\mathcal{A}\|_\diamond \|\mathcal{B}\|_\diamond$:

$$\|(\mathcal{N}_1 - \mathcal{I}) \circ (\mathcal{N}_2 - \mathcal{I})\|_\diamond \leq \|\mathcal{N}_1 - \mathcal{I}\|_\diamond \cdot \|\mathcal{N}_2 - \mathcal{I}\|_\diamond = \varepsilon_1 \varepsilon_2.$$

Substitute back to obtain

$$\|\mathcal{N}_1 \circ \mathcal{N}_2 - \mathcal{I}\|_\diamond \leq \varepsilon_1 + \varepsilon_2 + \varepsilon_1 \varepsilon_2. \quad \blacksquare$$

What is Fault torance computation?

Definition 6 (Physical Implementation Framework) Let $\mathcal{C} \subseteq \mathcal{H}$ be a quantum code with projector P , and U_L the ideal logical unitary decomposed as $U_L = U_t U_{t-1} \cdots U_1$.

A collection of physical subcircuits is a **physical implementation framework** if

1. **State preparation module:**

A noisy channel $\mathcal{P}$ generating encoded logical states $\rho = P\rho P$, with diamond norm error bound $\|\mathcal{P} - \mathcal{P}_L\|_\diamond \leq \varepsilon$;

2. **Encoded gate module:**

Composite channel $\mathcal{G}_k = \mathcal{N}_k \circ \mathcal{U}_k$, each single fault produces Kraus operators spanned by the correctable error set $\{E_i\}$;

$$\mathcal{F} = \mathcal{N}_g \circ \mathcal{U}_g \circ \mathcal{N}_{g-1} \circ \mathcal{U}_{g-1} \circ \cdots \circ \mathcal{N}_1 \circ \mathcal{U}_1,$$

3. **Syndrome measurement module:**

Projective measurement channel $\mathcal{M}_S$ for error diagnosis, its induced noise is correctable, and

$$\mathcal{R} \circ \mathcal{M}_S(\rho) \propto U_L \rho U_L^\dagger;$$

4. **Logical readout module:**

Final **projective measurement channel** $\mathcal{M}_L$, which maps encoded logical state to classical outcome with bounded physical error.

Theorem 7 (Quantum Fault-Tolerance Threshold Theorem1) Fix an $[[n, k, d]]$ stabilizer quantum error-correcting code with distance $d = 2t + 1$, which corrects all weight- t errors. Let all physical operations (qubit state preparation, unitary gates, syndrome measurement, readout, idle evolution) be CPTP channels $\mathcal{O}$. Let $\mathcal{I}$ denote the single-qubit identity channel, and define per-operation noise strength

$$\varepsilon = \|\mathcal{O} - \mathcal{I}\|_{\diamond}.$$

Assume all physical noise is independent single-qubit tensor noise: any n -qubit physical register evolves as $\bigotimes_{i=1}^n \mathcal{O}_i$.

There exists a universal positive constant threshold $\varepsilon_{th} > 0$ (depending only on the code distance d and choice of fault-tolerant gadgets) satisfying the following two statements:

1. If $\varepsilon < \varepsilon_{th}$: For any ideal noiseless quantum circuit $\mathcal{C}_{ideal}$ with L logical gates, any target logical error bound $\delta > 0$, there exists a fault-tolerant concatenated simulation circuit $\mathcal{C}_{ft}$ such that:
 - The logical channel implemented by $\mathcal{C}_{ft}$ satisfies

$$\|\mathcal{C}_{ft} - \mathcal{C}_{ideal}\|_{\diamond} \leq \delta;$$

- The total number of physical gates/qubits required scales as

$$\text{poly}(L, \log(1/\delta)),$$

i.e., overhead is polynomial in circuit length and polylogarithmic in target logical error.

- After k layers of concatenation, the effective logical noise satisfies

$$\varepsilon_k \leq C \cdot (\varepsilon/\varepsilon_{th})^{2^k}$$

for a constant $C > 0$ independent of k, ε . Logical error decays doubly exponentially with concatenation depth k .

2. If $\varepsilon > \varepsilon_{th}$: No finite number of concatenation layers can suppress logical noise to arbitrarily small δ ; logical error magnitude is bounded below by a constant independent of encoding overhead.

Proof

The proof proceeds in four core steps, relying on the diamond norm properties and Theorem 1.1 (independent tensor noise approximated by t -weight error channels).

Step 1: Fault-tolerant gadget single-layer error bound All elementary logical gadgets (transversal gate, fault-tolerant syndrome measurement, logical state preparation) are constructed to satisfy the key fault isolation property: any single physical fault induces at most one weight-1 error inside a code block, which is fully correctable by the distance- $2t + 1$ QECC. Decompose the gadgets total channel as ideal logical map plus noise residual:

$$\mathcal{G} = \mathcal{G}_{ideal} + \Delta\mathcal{G}.$$

By Theorem 1.1, independent single-qubit tensor noise with per-unit diamond error ε can be approximated by a t -qubit error map, with residual diamond error bounded by $\binom{n}{t+1}[(4e+2)\varepsilon]^{t+1}$. Only events with **two or more independent physical faults** produce uncorrectable high-weight errors contributing to logical noise. Single faults are eliminated by error correction. Thus the single-layer logical noise satisfies

$$\varepsilon_1 = \|\mathcal{G} - \mathcal{G}_{ideal}\|_{\diamond} \leq C_0 \varepsilon^2,$$

where $C_0 > 0$ is a constant only depending on code parameters and gadget structure.

Step 2: Concatenation recursion relation Define level- k logical noise ε_k as the diamond distance between level- k concatenated gadget and ideal logical operation. Level- k logical gadgets are built from many level- $k-1$ logical components. The same two-fault counting argument applies at the logical level: logical errors only arise when two or more level- $k-1$ logical faults occur simultaneously. Substitute the single-layer quadratic bound recursively:

$$\varepsilon_k \leq C_0 \cdot \varepsilon_{k-1}^2.$$

This is the core quadratic suppression recursion for concatenated codes.

Step 3: Threshold definition and decay analysis Define threshold value $\varepsilon_{\text{th}} = 1/\sqrt{C_0}$.

1. Case $\varepsilon < \varepsilon_{\text{th}}$: Then $C_0\varepsilon < 1$. Unfold recursion:

$$\begin{aligned} \varepsilon_1 &\leq C_0\varepsilon^2, \\ \varepsilon_2 &\leq C_0\varepsilon_1^2 \leq C_0 \cdot (C_0\varepsilon^2)^2 = C_0^{1+2}\varepsilon^{2^2}, \\ \varepsilon_k &\leq C_0^{2^k-1}\varepsilon^{2^k} = C \cdot \left(\frac{\varepsilon}{\varepsilon_{\text{th}}}\right)^{2^k}, \quad C = C_0^{-1}. \end{aligned}$$

The exponent 2^k grows doubly exponentially. For any target $\delta > 0$, we can choose finite k large enough such that $\varepsilon_k \leq \delta$. The required concatenation depth $k = O(\log \log(1/\delta))$, hence physical overhead $\text{poly}(L, \log(1/\delta))$.

2. Case $\varepsilon > \varepsilon_{\text{th}}$: Then $C_0\varepsilon > 1$. The recursion $\varepsilon_k \geq C_0\varepsilon_{k-1}^2$ forces logical noise to grow doubly exponentially with k . No amount of concatenation can drive logical error to zero; a uniform lower bound on $\|\mathcal{C}_{\text{ft}} - \mathcal{C}_{\text{ideal}}\|_{\diamond}$ exists for all k .

Step 4: Circuit composition extension Arbitrary long logical circuit $\mathcal{C}_{\text{ideal}}$ is a composition of many logical gadgets. Use diamond norm composition property

$$\|\mathcal{A}_1 \circ \mathcal{A}_2 - \mathcal{I}\|_{\diamond} \leq \|\mathcal{A}_1 - \mathcal{I}\|_{\diamond} + \|\mathcal{A}_2 - \mathcal{I}\| + \|\mathcal{A}_1 - \mathcal{I}\| \|\mathcal{A}_2 - \mathcal{I}\|.$$

Composing L logical gates each with noise ε_k yields total logical diamond error linear in $L \cdot \varepsilon_k$. By choosing sufficiently large k , we suppress $L\varepsilon_k \leq \delta$, completing the proof. ■

12.4 Plan for next week

- Prove the results

Theorem 8 *There is an exact sequence:*

$$1 \longrightarrow \mathcal{P}_n \xrightarrow{\lambda} \mathcal{C}_n \xrightarrow{\pi} \text{Sp}(2n, \mathbb{Z}_2) \longrightarrow 1.$$

- Study the group $\mathcal{C}_{\log, n} := \{\mu P_1 \otimes \dots \otimes P_n : P_i \in \langle X, Z, H \rangle^{\otimes n}\}$
- Read the book QECC-2026.
- Learn about type theory and Lean4.

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